

■ System Architecture & Circuit Connections

Transmitter Module (Diver) + Receiver Module (Surface Team)

System Overview

The system consists of two main units: the Transmitter Module worn by the scuba diver and the Receiver Module operated by the surface team. Health data is transmitted underwater using Li-Fi (light-based) communication.

Transmitter Module (Diver Unit)

Component	Connected To	Interface	Purpose
Temperature Sensor	ESP32 GPIO4	1-Wire	Body temperature reading
Pulse Oximeter (MAX30102)	ESP32 SDA/SCL	I2C	SpO2 and heart rate
Heartbeat Sensor	ESP32 ADC pin	Analog	Pulse rate detection
Panic Switch	ESP32 GPIO (pull-up)	Digital IN	Manual SOS trigger
LCD Display (16x2)	ESP32 via I2C	I2C	Local diver status display
Li-Fi Transmitter LED	ESP32 GPIO via transistor	Digital OUT	Optical data transmission
Relay Module	ESP32 GPIO	Digital OUT	Controls air motor
Air Motor	Relay NO contact	High power	Emergency balloon inflation
Buzzer (Diver)	ESP32 GPIO	Digital OUT	Local diver alarm
Power Supply	VIN / 5V pin of ESP32	DC 5V	Powers entire transmitter unit

Receiver Module (Surface Team Unit)

Component	Connected To	Interface	Purpose
Li-Fi Receiver (Photodiode)	ESP32 ADC / UART	Analog/Digital	Captures light signals from diver
ESP32 Receiver	All below components	Master controller	Decodes health data, triggers alerts
LCD Display (16x2)	ESP32 via I2C	I2C	Displays heart rate, SpO2, temperature
Buzzer (Surface)	ESP32 GPIO	Digital OUT	Audible alarm for abnormal vitals
Cloud (ThingSpeak)	ESP32 Wi-Fi	HTTP/MQTT	Remote data logging and dashboard
Power Supply	VIN / 5V pin of ESP32	DC 5V	Powers entire receiver unit

Li-Fi Communication Link

- Transmitter: Modulated LED light encodes sensor data (LED ON=1, LED OFF=0)

- Medium: Underwater visible light (blue-green wavelength preferred)
- Receiver: LDR/Photodiode detects light intensity changes, decodes data
- Range: 2–3 meters (line-of-sight required)
- Latency: <100 ms average
- Packet loss: Near-zero under clear water conditions
- Advantage over RF: No electromagnetic interference, works underwater

Power Supply Design

- Input: 230V AC mains
- Transformer: Steps down to 12-0-12V
- Bridge Rectifier (WM04): Converts AC to pulsating DC
- Capacitor (1000 μ F): Smoothing / reservoir capacitor
- 7805 IC: Produces regulated +5V for microcontroller and sensors
- LM317 IC: Produces regulated +3.8V for Li-Fi and analog circuits
- Protection: Current limiting + thermal overload built into 7805/LM317
- Indicator: LED + 220 Ω resistor as power-on pilot light

■ Safety Notes

- The air motor operates on high-power supply — handle relay wiring with care.
- Ensure all underwater components are properly waterproofed (IP67/IP68).
- Li-Fi transmitter LED must not shine directly into eyes at close range.
- MAX30102 sensor requires firm but gentle fingertip/wrist contact for accurate SpO2.
- Power supply must be isolated from water; use sealed battery pack for diver unit.